Chapter 1

The Ornstein–Uhlenbeck channel and Tweedie's identity

This chapter builds the forward process of the paper from nothing. It writes down the stochastic differential equation that adds the noise, solves it, and derives the three facts on which every later chapter rests: the noised law is a Gaussian convolution of the clean law; its per-site variance is one at every noise level; and its score is a rescaled posterior mean (Tweedie's identity). It ends with the observation protocol of the paper, one noised view per sequence on a twelve-level schedule, written out level by level. Throughout, $\mathbf{a} \in \mathbb{R}^L$ is a clean sequence with law P_0, $\mathbf{x}_t$ its noised version at level t with law P_t, and $\mathbf{z} \sim \mathcal{N}(0, \mathbf{I})$ a standard Gaussian vector independent of $\mathbf{a}$. In derivations we abbreviate $\alpha = e^{-t}$, as the paper's appendix does; in displayed statements we keep e^{-t}, as the paper's body does.

1.1 Stochastic differential equations in one paragraph

A **Brownian motion** $(W_t)_{t \geq 0}$ is a random continuous path with $W_0 = 0$ whose increments over disjoint intervals are independent and Gaussian, $W_t - W_s \sim \mathcal{N}(0, t - s)$ for $s < t$. Its formal time derivative $\eta(t) = \mathrm{d}W_t / \mathrm{d}t$ is **white noise**. It is **standardised** in the sense that $\mathbb{E}[\eta(t)\eta(t')] = \delta(t - t')$, which is the same as saying that $\int_s^t \eta(u)\,\mathrm{d}u = W_t - W_s$ has variance $t - s$. A **stochastic differential equation**

$$\mathrm{d}x_t = f(x_t, t)\ \mathrm{d}t + g(t)\ \mathrm{d}W_t \tag{1.1}$$

is shorthand for the recursion $x_{t+h} - x_t \approx f(x_t, t)\,h + g(t)\,(W_{t+h} - W_t)$ taken to the limit $h \to 0$; f is the **drift** (the deterministic push) and g the **diffusion coefficient** (the size of the random kick). The limit defines the Itô integral, and only two of its properties are needed here. First, for a deterministic integrand ϕ the random variable $\int_0^t \phi(u)\,\mathrm{d}W_u$ is Gaussian with mean zero and variance $\int_0^t \phi(u)^2\,\mathrm{d}u$ (the **Itô isometry**). Second, a linear equation can be solved with an integrating factor exactly as an ordinary differential equation can, because multiplying by a smooth deterministic function introduces no Itô correction term. In L dimensions W_t has L independent coordinates and Equation (1.1) is read coordinate by coordinate.

1.2 The variance-preserving Ornstein–Uhlenbeck process

Definition 1.1 (The channel)**.** The **Ornstein–Uhlenbeck channel** of the paper is the coordinatewise process

$$\mathrm{d}\mathbf{x}_t = -\mathbf{x}_t\ \mathrm{d}t + \sqrt{2}\ \mathrm{d}\mathbf{W}_t, \qquad \mathbf{x}_0 = \mathbf{a} \sim P_0, \tag{1.2}$$

with $\mathbf{W}$ an L-dimensional Brownian motion independent of $\mathbf{a}$. In white-noise form, $\mathrm{d}\mathbf{x}_t / \mathrm{d}t = -\mathbf{x}_t + \sqrt{2}\,\eta(t)$ with η standardised.

The drift $-\mathbf{x}_t$ pulls every coordinate towards zero at unit rate; the kick $\sqrt{2}\,\mathrm{d}\mathbf{W}_t$ is the same for every coordinate and independent across coordinates. Nothing couples the sites: the channel acts on each a_i separately, and this is what keeps the posterior factor graph a tree in the paper's Section 2.

Proposition 1.2 (Solution of the channel)**.** *For every $t \geq 0$,*

$$\mathbf{x}_t = e^{-t}\mathbf{a} + \sqrt{\Delta_t}\,\mathbf{z}, \qquad \Delta_t = 1 - e^{-2t}, \qquad \mathbf{z} \sim \mathcal{N}(0, \mathbf{I})\,, \tag{1.3}$$

with $\mathbf{z}$ independent of $\mathbf{a}$. Conditionally on $\mathbf{a}$, $\mathbf{x}_t \sim \mathcal{N}(e^{-t}\mathbf{a}, \Delta_t \mathbf{I})$.

Proof. Take one coordinate and drop the index. Put $y_t = e^t x_t$. The factor e^t is deterministic and smooth, so the ordinary product rule applies:

$$\mathrm{d}y_t = e^t x_t\,\mathrm{d}t + e^t\,\mathrm{d}x_t = e^t x_t\,\mathrm{d}t + e^t(-x_t\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}W_t) = \sqrt{2}\,e^t\,\mathrm{d}W_t.$$

Integrating from 0 to t gives $y_t = a + \sqrt{2}\int_0^t e^u\,\mathrm{d}W_u$, hence

$$x_t = e^{-t}a + \sqrt{2}\int_0^t e^{-(t-u)}\,\mathrm{d}W_u.$$

The integrand $\sqrt{2}\,e^{-(t-u)}$ is deterministic, so by the Itô isometry the integral is Gaussian with mean zero and variance

$$\int_0^t 2\,e^{-2(t-u)}\,\mathrm{d}u = 2\cdot\frac{1-e^{-2t}}{2} = 1 - e^{-2t} = \Delta_t.$$

It is independent of a because W is. Writing the integral as $\sqrt{\Delta_t}\,z$ with $z \sim \mathcal{N}(0,1)$ gives the scalar claim; the coordinates use independent Brownian motions, so the vector claim follows with $\mathbf{z} \sim \mathcal{N}(0, \mathbf{I})$. □

Remark 1.3 (Why $\sqrt{2}$). The variance $v_t = \mathrm{Var}(x_t)$ of one coordinate obeys $\mathrm{d}v_t/\,\mathrm{d}t = -2v_t + 2$, whose fixed point is $v = 1$. With the kick $\mathrm{d}W_t$ instead of $\sqrt{2}\,\mathrm{d}W_t$ the fixed point would be $\frac{1}{2}$ and one would have $\Delta_t = (1 - e^{-2t})/2$. The paper's convention (`paper/notation.tex`, item 4) fixes the noise as standardised and carries the $\sqrt{2}$ explicitly so that $\Delta_t = 1 - e^{-2t}$ throughout, in the forward and in the reverse equation. The variance-preserving SDE of Song and co-authors, $\mathrm{d}\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x}\,\mathrm{d}t + \sqrt{\beta(t)}\,\mathrm{d}\mathbf{W}$, is the same process when $\beta \equiv 2$; for a general schedule β the paper's t is $\frac{1}{2}\int_0^{t'}\beta(u)\,\mathrm{d}u$, and nothing below changes.

Remark 1.4 (The semigroup property). Noising for time t and then for time s is noising for time $t + s$: if $\mathbf{x}_t = e^{-t}\mathbf{a} + \sqrt{\Delta_t}\mathbf{z}$ and $\mathbf{x}_{t+s} = e^{-s}\mathbf{x}_t + \sqrt{\Delta_s}\mathbf{z}'$ with $\mathbf{z}'$ fresh, then $\mathbf{x}_{t+s} = e^{-(t+s)}\mathbf{a} + (e^{-s}\sqrt{\Delta_t}\mathbf{z} + \sqrt{\Delta_s}\mathbf{z}')$, and the bracket is Gaussian with variance $e^{-2s}(1 - e^{-2t}) + 1 - e^{-2s} = 1 - e^{-2(t+s)} = \Delta_{t+s}$ per coordinate. This is the Markov property of the channel, used in **??**.

1.3 Variance preservation and the Gaussian limit

Proposition 1.5 (Second moments of the noised law)**.** *Let $\mathbf{\Sigma}_0 = \mathrm{Cov}(\mathbf{a})$. Then $\mathbb{E}[\mathbf{x}_t] = e^{-t}\mathbb{E}[\mathbf{a}]$ and*

$$\mathrm{Cov}(\mathbf{x}_t) = e^{-2t}\mathbf{\Sigma}_0 + \Delta_t \mathbf{I} =: \mathbf{\Sigma}_t. \tag{1.4}$$

*In particular, if $\mathrm{Var}(a_i) = 1$ for every site, then $\mathrm{Var}(x_{t,i}) = e^{-2t} + \Delta_t = 1$ at every t: the channel **preserves the variance**.*

Proof. From Equation (1.3), $\mathbf{a}$ and $\mathbf{z}$ are independent, so covariances add: $\mathrm{Cov}(\mathbf{x}_t) = e^{-2t}\,\mathrm{Cov}(\mathbf{a}) + \Delta_t\,\mathrm{Cov}(\mathbf{z})$. □

The paper's chain (Section 2) has $a_1 \sim \mathcal{N}(0,1)$ and innovation variance $q = 1-\rho^2$, which gives $\mathrm{Var}(a_i) = 1$ and $\mathrm{Cov}(a_i, a_j) = \rho^{|i-j|}$ for every innovation law; hence every family shares the same $\mathbf{\Sigma}_0$ and the same $\mathbf{\Sigma}_t$.

Proposition 1.6 (The Gaussian limit)**.** *As $t \to \infty$, P_t converges in distribution to $\gamma = \mathcal{N}(0, \mathbf{I})$, whatever P_0 is, provided $\mathbb{E}|\mathbf{a}|^2 < \infty$.*

Proof. For a bounded continuous ϕ, $\mathbb{E}\phi(\mathbf{x}_t) = \mathbb{E}\phi(e^{-t}\mathbf{a} + \sqrt{\Delta_t}\mathbf{z})$. The term $e^{-t}\mathbf{a}$ tends to zero in probability by Chebyshev's inequality, $\mathbb{P}(e^{-t}|\mathbf{a}| > \epsilon) \le e^{-2t}\mathbb{E}|\mathbf{a}|^2/\epsilon^2$, and $\sqrt{\Delta_t}\mathbf{z} \to \mathbf{z}$ almost surely. Dominated convergence then gives $\mathbb{E}\phi(\mathbf{x}_t) \to \mathbb{E}\phi(\mathbf{z})$. □

The limit is the stationary law of the process: if $\mathbf{a} \sim \gamma$ then $\mathbf{x}_t \sim \mathcal{N}\big(0, e^{-2t}\mathbf{I} + \Delta_t \mathbf{I}\big) = \gamma$ for every t. The rate of convergence is the subject of **??**: the Hermite moment of degree m of P_t is exactly e^{-mt} times that of P_0, so the memory of P_0 fades order by order rather than all at once.

1.4 The noised law as a Gaussian convolution

Write γ_Δ for the density of $\mathcal{N}(0, \Delta\mathbf{I})$ on $\mathbb{R}^L$,

$$\gamma_\Delta(\mathbf{w}) = (2\pi\Delta)^{-L/2} \exp\Big(-\frac{|\mathbf{w}|^2}{2\Delta}\Big). \tag{1.5}$$

Proposition 1.7 (Density of the noised law)**.** *For every $t > 0$ the law P_t has the density*

$$P_t(\mathbf{x}) = \int_{\mathbb{R}^L} P_0(\,\mathrm{d}\mathbf{a})\, \gamma_{\Delta_t}(\mathbf{x} - e^{-t}\mathbf{a}) = \mathbb{E}_{P_0}\big[\gamma_{\Delta_t}(\mathbf{x} - e^{-t}\mathbf{a})\big], \tag{1.6}$$

which is strictly positive and infinitely differentiable in $\mathbf{x}$, whatever P_0 is (it need not have a density). Every derivative may be taken under the integral sign.

Proof. Conditionally on $\mathbf{a}$, $\mathbf{x}_t$ has density $\gamma_{\Delta_t}(\mathbf{x} - e^{-t}\mathbf{a})$ by Proposition 1.2; integrating over $\mathbf{a}$ gives Equation (1.6). Positivity is clear. For differentiation under the integral, every $\mathbf{x}$-derivative of $\gamma_{\Delta_t}(\mathbf{x} - e^{-t}\mathbf{a})$ is a polynomial in $(\mathbf{x} - e^{-t}\mathbf{a})/\Delta_t$ times the same Gaussian, and is therefore bounded uniformly in $\mathbf{a}$ on every bounded set of $\mathbf{x}$ by a constant, which is integrable against the probability measure P_0; dominated convergence applies. □

Two readings of Equation (1.6) are used. As an operator statement, P_t is the convolution of the dilated law (the law of $e^{-t}\mathbf{a}$) with the Gaussian γ_{Δ_t}. As a factor-graph statement, the integrand factorises over sites,

$$\gamma_{\Delta_t}(\mathbf{x} - e^{-t}\mathbf{a}) = \prod_{i=1}^{L} \ell_t(x_i \mid a_i), \qquad \ell_t(x_i \mid a_i) \propto \exp\Big[-\frac{(x_i - e^{-t}a_i)^2}{2\Delta_t}\Big], \tag{1.7}$$

so conditioning on $\mathbf{x}$ attaches one **unary likelihood factor** ℓ_t to each site and adds no edge among the latent variables. The code builds exactly this table: `log_likelihood_matrix` in `src/markov_score/noising.py` returns $\log \ell_t(x_i \mid u_k)$ on the quadrature grid, shifted per row by a constant so that no row underflows; the shift cancels in every normalised message and changes nothing.

1.5 The signal-to-noise ratio

Definition 1.8 (Signal-to-noise ratio)**.** The **signal-to-noise ratio** of the channel at level t is the ratio of the signal variance to the noise variance in one coordinate of Equation (1.3) for unit-variance data,

$$\mathrm{SNR}_t = \frac{e^{-2t}}{\Delta_t} = \frac{e^{-2t}}{1 - e^{-2t}} = \frac{1}{e^{2t} - 1}. \tag{1.8}$$

Table 1.1: The twelve-level schedule of the paper (`experiments/frozen_config.py`, $t_k = 0.05 \cdot r^{k-1}$ with $r = 60^{1/11} \approx 1.4509$), with the analytic values of the decay factor, the noise variance and the signal-to-noise ratio at each level. All entries are computed from the definitions; none is measured.

k	t_k	e^{-t_k}	Δ_{t_k}	SNR_{t_k}
1	0.0500	0.9512	0.0952	9.51
2	0.0725	0.9300	0.1350	6.40
3	0.1053	0.9001	0.1898	4.27
4	0.1527	0.8584	0.2632	2.80
5	0.2216	0.8012	0.3580	1.79
6	0.3215	0.7250	0.4743	1.11
7	0.4665	0.6272	0.6066	0.648
8	0.6769	0.5082	0.7417	0.348
9	0.9821	0.3745	0.8597	0.163
10	1.4250	0.2405	0.9422	0.061
11	2.0676	0.1265	0.9840	0.016
12	3.0000	0.0498	0.9975	0.0025

For small t, $\Delta_t = 2t + O(t^2)$ and $\mathrm{SNR}_t \approx 1/(2t)$, which diverges: the observation is almost clean. For large t, $\mathrm{SNR}_t \approx e^{-2t}$: the signal is a small perturbation of pure noise, and e^{-2t} is the natural small parameter of the expansion in **??**. The same quantity appears there under the name $b = e^{-2t}/\Delta_t$, together with $s = e^{-t}\mathbf{x}/\Delta_t$; the reader should keep in mind that b is the SNR. Inverting Equation (1.8), $t = \frac{1}{2}\log(1 + 1/\mathrm{SNR}_t)$, so the diffusion time is a logarithmic clock on the SNR.

1.6 Tweedie's identity

The **score** of the noised law is the gradient of its log-density, $S(\mathbf{x}, t) = \nabla_{\mathbf{x}} \log P_t(\mathbf{x})$. Tweedie's identity says that it is a rescaled residual of the posterior mean of the clean sequence. We write $\langle \mathbf{a} \rangle_{x,t} = \mathbb{E}[\mathbf{a} \mid \mathbf{x}_t = \mathbf{x}]$ for that posterior mean, in the bracket notation of the paper.

Theorem 1.9 (Tweedie's identity)**.** *For every $t > 0$ and every $\mathbf{x} \in \mathbb{R}^L$,*

$$S(\mathbf{x}, t) = \nabla_{\mathbf{x}} \log P_t(\mathbf{x}) = -\frac{\mathbf{x} - e^{-t}\langle \mathbf{a} \rangle_{x,t}}{\Delta_t}, \qquad \langle \mathbf{a} \rangle_{x,t} = \mathbb{E}[\mathbf{a} \mid \mathbf{x}_t = \mathbf{x}]. \tag{1.9}$$

Equivalently, $\langle \mathbf{a} \rangle_{x,t} = e^{t}\big(\mathbf{x} + \Delta_t\, S(\mathbf{x}, t)\big)$. No assumption on P_0 is needed beyond $\mathbb{E}|\mathbf{a}| < \infty$.

Proof. Scalar case first. By Proposition 1.7, $P_t(x) = \int P_0(\mathrm{d}a)\, \gamma_{\Delta_t}(x - \alpha a)$ with $\alpha = e^{-t}$, and the derivative may be taken inside:

$$\frac{\mathrm{d}}{\mathrm{d}x}\gamma_{\Delta_t}(x - \alpha a) = -\frac{x - \alpha a}{\Delta_t}\, \gamma_{\Delta_t}(x - \alpha a).$$

Hence

$$P_t'(x) = -\frac{1}{\Delta_t}\int P_0(\mathrm{d}a)\, (x - \alpha a)\, \gamma_{\Delta_t}(x - \alpha a) = -\frac{1}{\Delta_t}\Big[x\, P_t(x) - \alpha \int P_0(\mathrm{d}a)\, a\, \gamma_{\Delta_t}(x - \alpha a)\Big].$$

The posterior of a given $x_t = x$ is, by Bayes' rule, the probability measure $P_0(\mathrm{d}a)\, \gamma_{\Delta_t}(x - \alpha a)/P_t(x)$, so the last integral equals $P_t(x)\, \mathbb{E}[a \mid x_t = x]$. Dividing by $P_t(x) > 0$,

$$\frac{P_t'(x)}{P_t(x)} = -\frac{x - \alpha\, \mathbb{E}[a \mid x_t = x]}{\Delta_t}.$$

The integrability of $|a|$ under P_0 is what makes the last integral finite. In L dimensions, $\nabla_{\mathbf{x}}\gamma_{\Delta_t}(\mathbf{x}-\alpha\mathbf{a}) = -\Delta_t^{-1}(\mathbf{x}-\alpha\mathbf{a})\gamma_{\Delta_t}(\mathbf{x}-\alpha\mathbf{a})$ and the same three lines go through coordinate by coordinate. □

Remark 1.10 (The classical forms). Robbins (1956) wrote the identity for an empirical Bayes problem in which an observation $z \sim \mathcal{N}(\mu, \sigma^2)$ has an unknown prior on μ: $\mathbb{E}[\mu \mid z] = z + \sigma^2 (\log f)'(z)$ with f the marginal density of z. Efron (2011) named it Tweedie's formula. The channel is this setting with $\mu = e^{-t}\mathbf{a}$ and $\sigma^2 = \Delta_t$: $\mathbb{E}[e^{-t}\mathbf{a} \mid \mathbf{x}] = \mathbf{x} + \Delta_t S(\mathbf{x}, t)$, which is the second form in Theorem 1.9. The code uses it in both directions: `score_from_mean` in `src/markov_score/theory.py` turns a posterior mean into a score, and `denoising_readout` in `src/markov_score/reverse.py` turns a score into a posterior mean.

Proposition 1.11 (Second-order Tweedie identity)**.** *For $t > 0$ the posterior covariance is also a function of the noised law:*

$$\operatorname{Cov}(\mathbf{a} \mid \mathbf{x}_t = \mathbf{x}) = \frac{\Delta_t}{e^{-2t}}\Big(\mathbf{I} + \Delta_t \nabla_{\mathbf{x}}^2 \log P_t(\mathbf{x})\Big), \tag{1.10}$$

so that $\operatorname{div} S(\mathbf{x}, t) = \operatorname{tr} \nabla^2 \log P_t(\mathbf{x}) = -L/\Delta_t + (e^{-2t}/\Delta_t^2) \operatorname{tr} \operatorname{Cov}(\mathbf{a} \mid \mathbf{x}_t = \mathbf{x})$.

Proof. Differentiate Equation (1.9) once more. With $m(\mathbf{x}) = \langle \mathbf{a} \rangle_{x,t}$, $\nabla^2 \log P_t = -\mathbf{I}/\Delta_t + (\alpha/\Delta_t)\nabla_{\mathbf{x}} m$. Writing $m_i(\mathbf{x}) = \int a_i \, P_0(\mathrm{d}\mathbf{a}) \gamma_{\Delta_t}(\mathbf{x} - \alpha\mathbf{a})/P_t(\mathbf{x})$ and differentiating the quotient with respect to x_j,

$$\partial_j m_i = -\frac{1}{\Delta_t}\Big(\mathbb{E}[a_i(x_j - \alpha a_j) \mid \mathbf{x}] - m_i\,(x_j - \alpha m_j)\Big) = \frac{\alpha}{\Delta_t}\big(\mathbb{E}[a_i a_j \mid \mathbf{x}] - m_i m_j\big) = \frac{\alpha}{\Delta_t} \operatorname{Cov}(a_i, a_j \mid \mathbf{x}),$$

because the terms in x_j cancel. Substituting gives $\nabla^2 \log P_t = -\mathbf{I}/\Delta_t + (\alpha^2/\Delta_t^2) \operatorname{Cov}(\mathbf{a} \mid \mathbf{x})$, which is Equation (1.10) rearranged. □

Why it matters here. The divergence of the score is what implicit score matching (**??**) has to compute, and Equation (1.10) says what it is: up to constants, the trace of the posterior covariance. It also shows that the score is a gradient field whose Jacobian is symmetric, a property a network's output need not have.

Corollary 1.12 (The Gaussian case)**.** *If $P_0 = \mathcal{N}(0, \mathbf{\Sigma}_0)$ then $P_t = \mathcal{N}(0, \mathbf{\Sigma}_t)$ with $\mathbf{\Sigma}_t$ as in Equation* (1.4)*, the score is linear, $S(\mathbf{x}, t) = -\mathbf{\Sigma}_t^{-1}\mathbf{x}$, and the posterior mean is $\langle \mathbf{a} \rangle_{x,t} = e^{-t}\mathbf{\Sigma}_0\mathbf{\Sigma}_t^{-1}\mathbf{x}$.*

Proof. A Gaussian pushed through Equation (1.3) stays Gaussian, with the covariance of Proposition 1.5. Then $\log P_t(\mathbf{x}) = -\frac{1}{2}\mathbf{x}^\top \mathbf{\Sigma}_t^{-1}\mathbf{x} + \text{const}$ and $S = -\mathbf{\Sigma}_t^{-1}\mathbf{x}$. Inverting Equation (1.9),

$$\langle \mathbf{a} \rangle_{x,t} = \frac{\mathbf{x} + \Delta_t S}{\alpha} = \frac{(\mathbf{I} - \Delta_t \mathbf{\Sigma}_t^{-1})\mathbf{x}}{\alpha} = \frac{(\mathbf{\Sigma}_t - \Delta_t \mathbf{I})\mathbf{\Sigma}_t^{-1}\mathbf{x}}{\alpha} = \frac{\alpha^2 \mathbf{\Sigma}_0 \mathbf{\Sigma}_t^{-1}\mathbf{x}}{\alpha} = \alpha\, \mathbf{\Sigma}_0 \mathbf{\Sigma}_t^{-1}\mathbf{x}. \qquad \square$$

This is the **linear minimum-mean-square estimator** (LMMSE) of $\mathbf{a}$ from $\mathbf{x}$, and the paper's Proposition 1 (Section 2) shows that it is exactly what Gaussian message closure computes for *any* innovation law with the same $\mathbf{\Sigma}_0$. The score $S_{\mathrm{G}}(\mathbf{x}, t) = -\mathbf{\Sigma}_t^{-1}\mathbf{x}$ of this **covariance twin** is the reference against which the paper measures the price of second-order closure (**??**).

1.7 The posterior mean as a denoiser

Proposition 1.13 (The posterior mean is the best denoiser)**.** *For every measurable $f \colon \mathbb{R}^L \to \mathbb{R}^L$ with $\mathbb{E}|f(\mathbf{x}_t)|^2 < \infty$,*

$$\mathbb{E}\big|\mathbf{a} - f(\mathbf{x}_t)\big|^2 = \mathbb{E}\big|\mathbf{a} - \langle \mathbf{a} \rangle_{x,t}\big|^2 + \mathbb{E}\big|\langle \mathbf{a} \rangle_{x,t} - f(\mathbf{x}_t)\big|^2. \tag{1.11}$$

The first term on the right, written MMSE_t*, is the smallest mean squared error any denoiser can reach; it is reached only by $f = \langle \mathbf{a} \rangle_{x,t}$.*

Proof. Expand $|\mathbf{a}-f|^2 = |(\mathbf{a}-m)+(m-f)|^2$ with $m = \langle \mathbf{a} \rangle_{x,t}$. The cross term is $2\,\mathbb{E}\langle \mathbf{a}-m,\, m-f\rangle = 2\,\mathbb{E}\langle \mathbb{E}[\mathbf{a}-m \mid \mathbf{x}_t],\, m-f\rangle = 0$, because $m-f$ is a function of $\mathbf{x}_t$ and $\mathbb{E}[\mathbf{a}-m \mid \mathbf{x}_t] = 0$. □

Two consequences organise the paper's measurements. First, any estimate $\widehat{m}$ of the posterior mean converts into an estimate of the score and the errors are proportional:

$$\widehat{S} = -\frac{\mathbf{x} - e^{-t}\widehat{m}}{\Delta_t} \quad \Longrightarrow \quad \widehat{S} - S = \frac{e^{-t}}{\Delta_t}(\widehat{m} - \langle \mathbf{a} \rangle_{x,t}), \tag{1.12}$$

an exact identity that `score_mean_identity_residual` in `src/markov_score/metrics.py` checks to machine precision. Every estimator in the paper, the tree-exact recursion, the fitted kernel, the LMMSE twin and the networks, produces a posterior mean and is scored through Equation (1.12). Second, the energy of the score is tied to the MMSE.

Proposition 1.14 (Score energy)**.** $\mathbb{E}_{P_t}|S(\mathbf{x}_t, t)|^2 = L/\Delta_t - (e^{-2t}/\Delta_t^2)\,\mathrm{MMSE}_t$.

Proof. By Equation (1.3), $-(\mathbf{x}_t - e^{-t}\mathbf{a})/\Delta_t = -\mathbf{z}/\sqrt{\Delta_t}$, and by Theorem 1.9 the score is its conditional expectation, $S(\mathbf{x}_t, t) = \mathbb{E}[-\mathbf{z}/\sqrt{\Delta_t} \mid \mathbf{x}_t]$. Apply Equation (1.11) to the random vector $-\mathbf{z}/\sqrt{\Delta_t}$ in place of $\mathbf{a}$ and $f = S$: $\mathbb{E}|\mathbf{z}|^2/\Delta_t = \mathbb{E}|S|^2 + \mathbb{E}|-\mathbf{z}/\sqrt{\Delta_t} - S|^2$. The last term is $\mathbb{E}|\, e^{-t}(\mathbf{a} - \langle \mathbf{a} \rangle_{x,t})/\Delta_t\,|^2 = (e^{-2t}/\Delta_t^2)\mathrm{MMSE}_t$ by Equation (1.12) applied to $\widehat{m} = \mathbf{a}$; and $\mathbb{E}|\mathbf{z}|^2 = L$. □

At low noise the energy is close to L/Δ_t and large; at high noise it tends to L, the energy of the score of γ. This is why the paper's pooled estimand (**??**) is dominated by its lowest levels.

1.8 One noised view per sequence and the twelve-level schedule

The paper observes M sequences $\mathbf{a}^1, \ldots, \mathbf{a}^M$ and, in its own estimator's interface (interface C of Section 5), sees each of them exactly once, at one level t_μ, as $\mathbf{x}^\mu = e^{-t_\mu}\mathbf{a}^\mu + \sqrt{\Delta_{t_\mu}}\mathbf{z}^\mu$; the clean sequence, a second noised view, and the noise draw itself are never seen. The sequences are split evenly across the twelve levels of Table 1.1, so each level receives $M/12$ views (`_one_view_groups` in `experiments/e03_baselines.py`; `one_view=True` in `experiments/protocol.py`). The likelihood of one view is then $P_{t_\mu,\theta}(\mathbf{x}^\mu)$, the density of Equation (1.6) under the parameterised kernel, and the marginal log-likelihood of the paper's Section 4 is the sum over views. The schedule is log-spaced so that the eleven ratios t_{k+1}/t_k are equal; on the SNR clock of Equation (1.8) the levels run from an almost clean observation (level 1) to almost pure noise (level 12), and the crossover scale t_* of **??** at the frozen correlation lies between the ninth and the tenth level. The networks of the paper draw the level of each training example uniformly from the same twelve values (`t_levels` in `src/markov_score/nn_torch.py`), so that their population objective is an average over the schedule (**??**).

1.9 Worked example

Example 1.15 (A two-point prior at $t = \log 2$)**.** Let $L = 1$ and let $a = \pm 1$ with probability $\frac{1}{2}$ each, so that $\mathrm{Var}(a) = 1$. At $t = \log 2$ we have $e^{-t} = \frac{1}{2}$, $\Delta_t = 1 - \frac{1}{4} = \frac{3}{4}$ and $\mathrm{SNR} = \frac{1}{3}$. By Equation (1.6),

$$P_t(x) = \tfrac{1}{2}\gamma_{3/4}(x - \tfrac{1}{2}) + \tfrac{1}{2}\gamma_{3/4}(x + \tfrac{1}{2}).$$

The posterior of a given x puts mass proportional to $\exp[-(x \mp \frac{1}{2})^2/(2 \cdot \frac{3}{4})]$ on ± 1; the ratio of the two weights is $\exp(2 \cdot \frac{1}{2}x/\frac{3}{4}) = \exp(4x/3)$, so

$$\langle a \rangle_{x,t} = \tanh(\tfrac{2}{3}x), \qquad S(x,t) = -\frac{x - \frac{1}{2}\tanh(\frac{2}{3}x)}{3/4} = -\tfrac{4}{3}\Big(x - \tfrac{1}{2}\tanh\tfrac{2x}{3}\Big).$$

At $x = 1$: $e^{4/3} = 3.7937$, so $\tanh(2/3) = (3.7937 - 1)/(3.7937 + 1) = 0.5828$, $\langle a \rangle_{x,t} = 0.5828$ and $S(1,t) = -\frac{4}{3}(1 - 0.2914) = -0.9448$. The covariance twin of this prior is $\mathcal{N}(0,1)$, whose noised law is again $\mathcal{N}(0,1)$; its score at $x = 1$ is $S_{\mathrm{G}} = -1$ and its posterior mean is $e^{-t}x = 0.5$. The true denoiser pulls harder towards $+1$ than the linear one, by 0.083, and the score gap $S - S_{\mathrm{G}} = +0.0552$ is, by Equation (1.12), exactly $(e^{-t}/\Delta_t)(0.5828 - 0.5) = \frac{2}{3} \cdot 0.0828$. As a check of Corollary 1.12 on the twin: $-(x - \frac{1}{2} \cdot \frac{1}{2}x)/\frac{3}{4} = -x$, as it must be.

Example 1.16 (Reading the schedule)**.** At level 8 of Table 1.1, $t = 0.6769$, a site with clean value $a_i = 2$ is observed as $x_i = 0.5082 \cdot 2 + 0.8612\, z_i = 1.016 + 0.861\, z_i$: the signal has half its amplitude and the noise has standard deviation $\sqrt{0.7417} = 0.861$. At level 12 the same site is observed as $0.0996 + 0.9988\, z_i$; its amplitude is one twentieth and the SNR is $1/400$. The information a single view carries about the site's fourth cumulant is, by the paper's Theorem 3, proportional to e^{-8t}, which between these two levels falls by the factor $e^{-8(3-0.6769)} = e^{-18.58} \approx 8.5 \times 10^{-9}$.

Where it is used. The channel is equation (2) of the paper (Section 2, "Data and channel") and Tweedie's identity is the first sentence of "The score is a posterior mean"; the Gaussian corollary is the covariance twin of the same section and of Proposition 1. The twelve-level schedule is the frozen configuration of Appendix C, and the one-view interface is interface C of Section 5. Code: the channel and its likelihood table are `src/markov_score/noising.py`; the LMMSE mean and score and the mean-to-score conversion are in `src/markov_score/theory.py`; the score-to-mean readout is in `src/markov_score/reverse.py`; the schedule is `t_grid` in `experiments/frozen_config.py` and the one-view flag is in `experiments/protocol.py`. Experiments: every registered experiment E01–E21 noises its data through the two lines of Equation (1.3).

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